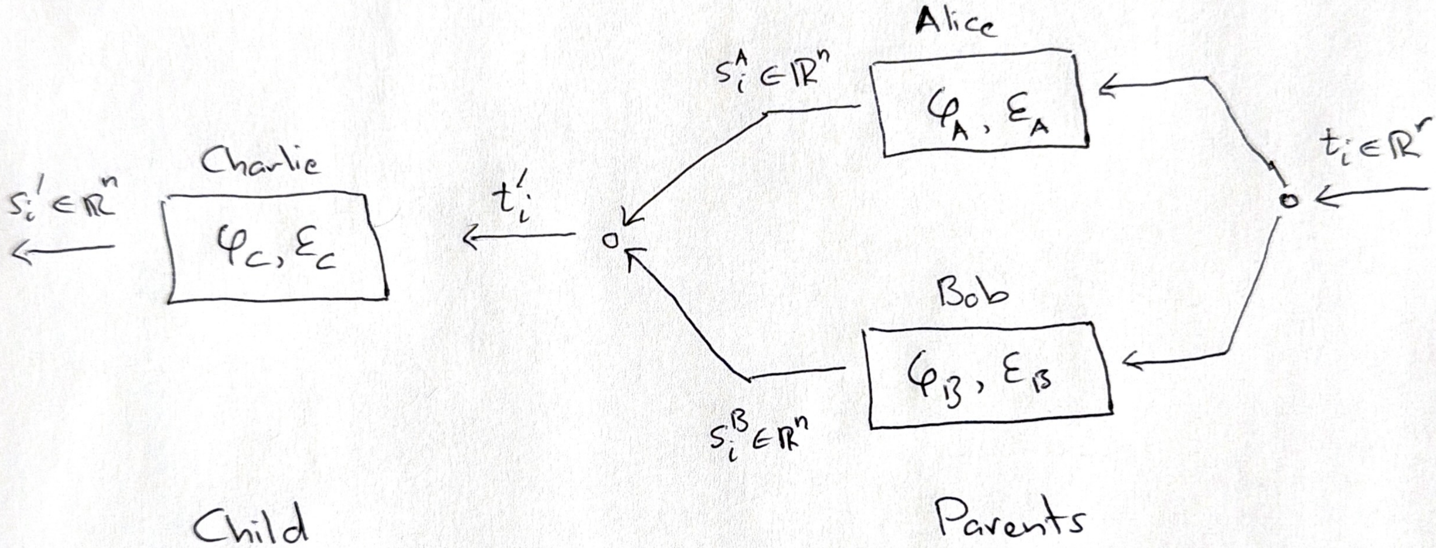
Further Noken genetic operators:

Let $u = (u_1, u_2, \ldots, u_n)^T$, $v = (v_1, v_2, \ldots, v_n)^T \in \mathbb{R}^n$

$$u \otimes v = (u_1 v_1, \ldots, u_1 v_n, \ldots, u_n v_1, \ldots, u_n v_n)^T$$

$$u \circ v = (u_1 v_1, u_2 v_2, \ldots, u_n v_n)^T$$



$$\varphi_C, \varepsilon_C = \begin{cases} \varphi_A, \varepsilon_A & \text{if } \varepsilon_A < \varepsilon_B \\ \varphi_B, \varepsilon_B & \text{otherwise} \end{cases}$$

Kronecker

$$t_i' \triangleq s_i^A \otimes s_i^B$$

Hadamard

$$t_i' \triangleq s_i^A \circ s_i^B$$

Let $u = (u_0, \ldots, u_{n-1})^T$, $v = (v_0, \ldots, v_{n-1})^T \in \mathbb{R}^n$

For $k = 0, 1, \ldots, n-1$

define $(u * v)_k = \sum_{i+j=k} u_i v_j$

Convolution

$$t'_i \triangleq s_i^A * s_i^B$$

Operator genetic operations

Rank-one operator

$$(v \otimes u)\, w \triangleq v \langle u, w \rangle$$

$$t_i' \triangleq (s_i^A \otimes s_i^B)\, t_i = s_i^A \langle s_i^B, t_i \rangle$$

Diagonal - Circulant operators

$$\text{circulant } C(c) \triangleq \begin{bmatrix} c_1 & \cdot & \cdot & c_{n-1} & c_n \\ c_n & c_1 & \cdot & \cdot & c_{n-1} \\ \vdots & \cdot & \cdot & & \vdots \\ & & & \cdot & \\ c_2 & \cdot & \cdot & c_n & c_1 \end{bmatrix}$$

$$\text{diagonal } D(c) \triangleq \begin{bmatrix} c_1 & & & 0 \\ & c_2 & & \\ & & \ddots & \\ 0 & & & c_n \end{bmatrix}$$

gates p, q, x, y

filters g, h

input v

Hyena

$$t' = D(x^N)\,C(h^N)\ldots D(x^1)\,C(h^1)\,v$$

TriAd

$$t' = [D(x^1)\,C(h^1)\,D(y^1) + \ldots + D(x^N)\,C(h^N)\,D(y^N)]\,v$$

Centipede

$$t' = T_N \ldots T_2\,T_1\,v$$

where $T_n = [D(p^n)\,C(g^n)\,D(q^n) + D(x^n)\,C(h^n)\,D(y^n)]$

Noken genetic operations are defined by token or noken filters and free ~~param~~ parameter gates, and token or noken input, eg $t'_i = D(x)\,C(s^A_j)\,t_i$, $j \neq i$, $x \propto \frac{1}{E_A}$

The Krylov decomposition of low displacement rank matrices ensures fast computation of these Noken genetic operations. [See: 'Structured Transforms for Small-Footprint Deep Learning' - Sindhwani, Sainath and Kumar, arXiv: 1510.01722v1 [stat.ML] 6 Oct 2015.]